Number Sequences & Patterns

Year 4 · **Homework Worksheet**

Worked Example

Worked example: Find the next two terms: 6, 12, 18, 24, __, __.

Answer: Add 6 each time. Next terms: 30, 36.

Easy / Standard (15 questions)

1. Find the next two terms: 4, 8, 12, 16, __, __.
2. Find the next two terms: 20, 40, 60, __, __.
3. Which number continues 5, 10, 15, 20, __? A) 22 B) 24 C) 25 D) 30
4. Find the missing term: 6, __, 18, 24, 30.
5. Find the next two terms: 90, 80, 70, __, __.
6. Find the next two terms: 1, 4, 7, 10, __, __.
7. Select ALL the sequences that count up in steps of 3: A) 3,6,9,12 B) 2,5,8,11 C) 15,18,21,24 D) 4,8,12,16 E) 9,12,15,18
8. Find the missing term: 8, 16, __, 32, 40.
9. Find the next two terms: 60, 54, 48, __, __.
10. Which is the rule for 7, 14, 21, 28…? A) add 5 B) add 7 C) add 9 D) multiply by 3
11. Find the next two terms: 3, 8, 13, 18, __, __.
12. Find the missing term: 40, 35, __, 25, 20.
13. Select ALL the numbers below that could continue the sequence 8, 16, 24, 32…: A) 36 B) 40 C) 44 D) 48 E) 50
14. Find the next two terms: 9, 18, 27, __, __.
15. Which number is missing? 14, 18, 22, __, 30. A) 24 B) 25 C) 26 D) 27

Standard (15 questions)

1. Find the next two terms: 3, 6, 12, 24, __, __.
2. Find the next two terms: 1, 3, 9, 27, __, __.
3. What is the rule for 5, 10, 20, 40…? A) add 5 B) add 10 C) double each time D) add 20
4. Find the missing term: 3, 7, __, 15, 19.
5. Find the next two terms: 160, 80, 40, __, __.
6. A pattern of hexagons has 6 sides in shape 1, 10 sides in shape 2 (joined), 14 sides in shape 3. How many sides in shape 4?
7. Select ALL the sequences below that step up in 6s: A) 6,12,18,24 B) 4,9,14,19 C) 10,16,22,28 D) 3,10,17,24 E) 8,14,20,26
8. Find the missing term: 11, 14, __, 20, 23.
9. Find the next two terms: 85, 75, 65, __, __.

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10. Which is the missing term? 36, 30, 24, __, 12. A) 16 B) 18 C) 20 D) 22

11. Find the next two terms: 12, 24, 36, 48, __, __.

12. A pattern of dots triples each time: 2, 6, 18, 54, __. What is the next term?

13. Find the missing term: 90, 77, __, 51, 38.

14. Which rule describes 4, 9, 16, 25...? A) add consecutive odd numbers starting at 5 B) add 5 each time C) multiply by 2 D) add 9 each time

15. Select ALL the numbers below that are square numbers: A) 16 B) 20 C) 36 D) 40 E) 49

Standard / Advanced (15 questions)

1. A sequence starts at 4 and adds 3 each time. What is the 5th term?

2. The rule for a sequence is 'double then subtract 2', starting at 3. What is the 3rd term?

3. A sequence starts at 2 and adds 5 each time. Which of these is the 4th term? A) 12 B) 17 C) 20 D) 22

4. A sequence starts at 1 and adds 2 each time (so it goes 1, 3, 5, 7...). Select ALL of these that are terms in the sequence: A) 7 B) 8 C) 15 D) 20 E) 21

5. A stadium has 10 seats in row 1, and each row behind has 3 more seats than the row in front. How many seats are in row 5?

6. A sequence begins at 50 and decreases by 8 each time. Which is the first negative term, and what is its value?

7. A tiler lays rows of tiles: row 1 has 4 tiles, row 2 has 7 tiles, row 3 has 10 tiles, each row 3 more than the last. How many tiles are needed for the first 3 rows altogether?

8. A sequence starts at 3 and adds 4 each time. Is the 10th term equal to 39? A) Yes, because 3 + 4×9 = 39 B) No, the correct answer is 43 C) No, the correct answer is 37 D) No, the correct answer is 40

9. A savings jar starts with £5 and gains £4 every week. After how many weeks will it first contain more than £30?

10. Two sequences are described: Sequence A starts at 3 and adds 5 each time. Sequence B starts at 11 and adds 3 each time. Write out the first 5 terms of each and say whether they ever show the same number.

11. A pattern of square tiles uses matchsticks: 1 square uses 4 sticks, and each extra square added in a row needs 3 more sticks. How many sticks are needed for a row of 6 squares?

12. A sequence starts at 6 and adds 5 each time (6, 11, 16, 21...). Which statement is TRUE? A) The 10th term is 51 B) The 10th term is 56 C) The sequence contains the number 50 D) The sequence contains the number 100

13. A pattern of dots forms triangular numbers 1, 3, 6, 10, 15... A pupil needs 21 dots for a triangular pattern. Which triangular number is 21?

14. A sequence starts at 7 and adds 4 each time (7, 11, 15, 19...). Select ALL of the following that are TRUE: A) The common difference is 4 B) The 6th term is 27 C) The sequence contains the number 51 D) The sequence contains the number 31 E) The 6th term is 31

15. A function machine adds 3, then doubles the result. If the input is 4, what is the output?

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